

Chapter 5 : Data Compression.

Def A source code $C: X \rightarrow \mathcal{D}^*$

X = random V.

X = sample space.

$$\mathcal{D} = \{0, 1, \dots, D-1\}$$

ex) $D=2$. $\mathcal{D} = \{0, 1\}$. $\mathcal{D}^* = \{01, 000, 101, 111, \dots\}$

• Expected length of C : $EL(X) = \sum_{x \in X} p(x) l(x) =: L(C)$

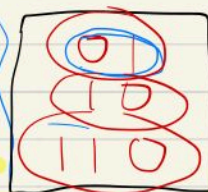
Instantaneous Code
(prefix)

Uniquely decodable

Nonsingular

All Code

x	C
a	01
b	11
c	111



0110110101

a	10
b	01
c	110
d	010
e	10

Thm (Kraft inequality).

Instantaneous code

$$\sum_i D^{-l_i} \leq 1$$

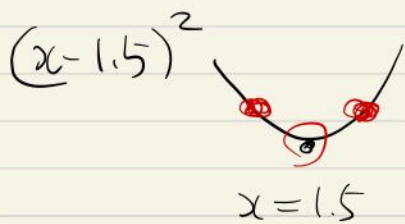
construct.

(McMillan inequality)

Uniquely Decodable codes : The same situation holds.

min $El(X) = L(C) = \sum_i p_i l_i$
 subject to $\sum_i D^{-l_i} \leq 1$, $l_i \in \mathbb{N}$

Indeed $l_i^* = -\log_D p_i \Rightarrow$ global minimum.



$p_i = D^{-l_i}$
 D -adic

$-\sum p_i \log_D p_i$
 $H_D(X)$

(Thm) [Not that too far away]

$l_1^*, l_2^*, \dots, l_m^*$ optimal codeword lengths for a D -ary alphabet. $L^* = \sum p_i l_i^*$

$H_D(X) \leq L^* < H_D(X) + 1$

$l(x) = \lceil \log_{D(x)} \frac{1}{p(x)} \rceil$ ($\lceil 2.8 \rceil = 3$, $\lceil 1.5 \rceil = 2$)
 Shannon Code.

Interest: Optimal & instantaneous prefix.

Huffman Code.

EX (Wikipedia)

A-EADD-DAD-CEDED-A-BAD-BABE-A-BEADED-ABACA-BED

$$\begin{array}{lcl}
 \textcircled{1} & A: 11 & \Rightarrow \textcircled{2} \\
 & B: 6 & \left. \begin{array}{l} C: 2 \\ B: 6 \end{array} \right\} CB = 8 \\
 & C: 2 & E: 7 \\
 & D: 10 & -: 10 \\
 & E: 7 & D: 10 \\
 & -: 10 & A: 11
 \end{array}$$

$$\begin{array}{lcl}
 \Rightarrow & E = 7 & ECR = 15 \\
 & CB = 8 & \\
 & - = 10 & \\
 & D = 10 & \\
 & A = 11 &
 \end{array}$$

$$(P_i) \times (Q_i)$$

10 00 110 100 101

A	0	10
B	00	1111
C	000	1110
D	0000	01
E	0001	110
-	0010	00

Existence of Optimal Code? $\Rightarrow$ Yes



Uniqueness of Optimal Code $\Rightarrow$ No.

$\therefore$ can make optimal codes many

$\Downarrow$
How to find it?

$\begin{array}{c} 10 \\ 01 \\ 111 \end{array} \rightsquigarrow \begin{array}{c} 01 \\ 10 \\ 000 \end{array}$

$\therefore$ Well, Huffman Code.

[Thm] (Huffman Coding is Optimal)

(pf) Use induction on # of symbols.

Assume $p = (p_1, p_2, \dots, p_m)$ $p_1 \geq p_2 \geq \dots \geq p_m$.

$\Downarrow$ condense

$p' = (p_1, p_2, \dots, p_{m-2}, p_{m-1} + p_m)$

[Def: Canonical Optimal Code]

1. $p_i > p_j \Rightarrow l_i \leq l_j$

2. Two longest codewords

① have the same lengths

② differ only in the last bit

③ correspond to least likely symbols.

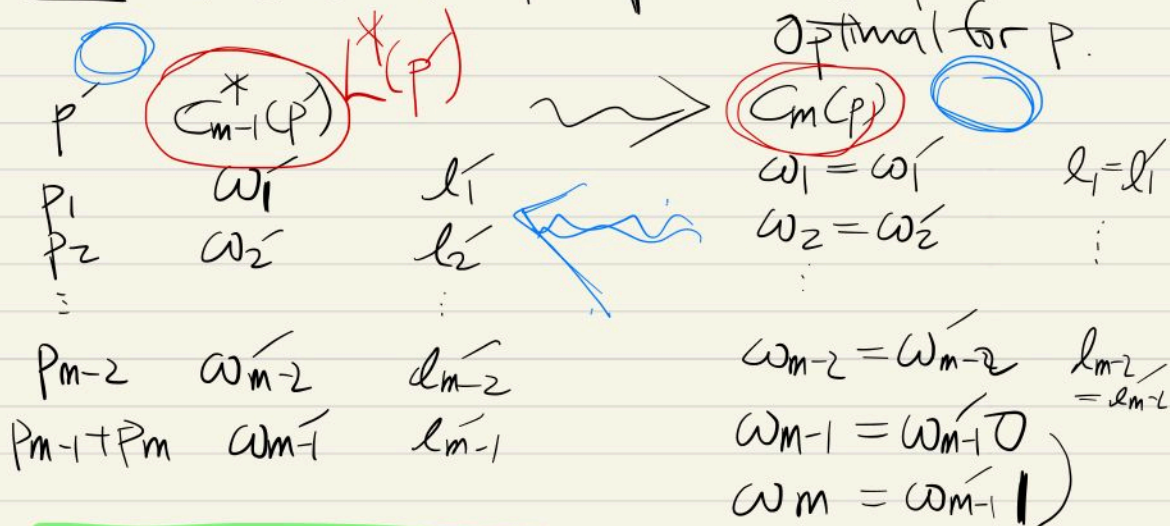
11100

11101

We have $C_m^*(p)$: Canonical Optimal Code.

$C_{m-1}^*(p')$: Optimal Code.

Claim Extension of Optimal code for p' is optimal for p .



$$L(p) = L^*(p) + p_{m-1} + p_m$$

$$l_{m-1} = l_{m-1}' + 1$$

$$l_m = l_{m-1}' + 1$$

Now from $C_m^*(p)$ construct a code p' by Huffman reduction.

$$L(p') = \sum_{i=1}^{m-2} p_i l_i + p_{m-1} (l_{m-1}' - 1) + p_m (l_{m-1}' - 1)$$

$$= L^*(p) - p_{m-1} - p_m$$

$$\Rightarrow L(p) - L^*(p') = L^*(p) - L(p')$$

$$(\underbrace{L(p) - L^*(p)}_{\geq 0}) + (\underbrace{L(p') - L^*(p')}_{\geq 0}) = 0$$

$$L(p) = L^*(p)$$

Shannon-Fano-Elias Coding

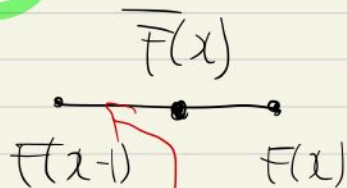
$$F(x) = \sum_{i \leq x} p(i) \rightsquigarrow \text{modify } \bar{F}(x) = \sum_{i \leq x} p(i) + \frac{1}{2}p(x)$$

$$\rightsquigarrow \lfloor \bar{F}(x) \rfloor_{l(x)} \quad (\text{truncated } \bar{F})$$

e.g. $\bar{F} = 0.110101001$

$l(x) = 5$

$\lfloor \bar{F} \rfloor_5 = 0.11010$



$$l(x) = \lceil \log \frac{1}{p(x)} \rceil + 1$$

$$\Rightarrow F(x-1) < \lfloor \bar{F}(x) \rfloor_{l(x)} < \bar{F}(x)$$

• Notion of prefix-free

Given $\lfloor \bar{F}(x) \rfloor_{l(x)} = 0.z_1 z_2 \dots z_{l(x)}$

$$\begin{cases} 0.1101010 \\ 0.1101011 \end{cases}$$

$$\longleftrightarrow [0.z_1 \dots z_{l(x)}, 0.z_1 \dots z_{l(x)} + \frac{1}{2^{l(x)}})$$

prefix-free code

$\iff$ such intervals are disjoint.

Shannon Code. $\ell(x) = \lceil \log \frac{1}{p(x)} \rceil$

[Thm] In Shannon Code, $\ell'(x)$: any uniquely decodable code length

$$\Rightarrow \Pr(\ell(X) \geq \ell'(X) + c)$$
$$\leq \frac{1}{2^{c-1}}$$

i.e. " $\ell(X)$ is quite short"

$$\Pr(\lceil \log \frac{1}{p(X)} \rceil \geq \ell'(X) + c)$$

$$\leq \Pr(\lceil \log \frac{1}{p(X)} \rceil \geq \ell'(X) + c)$$

$$= \Pr(p(X) \leq 2^{-\ell'(X) - c + 1})$$

$$= \sum_{p(X) \leq 2^{-\ell'(X) - c + 1}} p(X) \leq \sum_{p(X) \leq 2^{-\ell'(X) - c + 1}} 2^{-\ell'(X) - c + 1}$$

$$\leq \sum_x 2^{-\ell'(x) - c + 1}$$

$$\leq 2^{-(c-1)}$$

□

Thm $p(x) = \text{dyadic mass function}$

$$l(x) = \log_2 \frac{1}{p(x)}, \quad \text{prob. } l' : \text{other uniquely decodable binary code.}$$

Then, $\Pr(l(x) \leq l'(x))$
 $\geq \Pr(l(x) > l'(x))$

(pf) $\Pr(l < l') - \Pr(l > l')$
 $= \sum_{l < l'} p(x) - \sum_{l > l'} p(x)$

$$\text{sgn}(x) = \begin{cases} 1 & x > 0 \\ 0 & x = 0 \\ -1 & x < 0 \end{cases}$$

$$= \sum_x p(x) \text{sgn}(l(x) - l'(x)) \quad (*)$$

Note $\text{sgn}(x) \leq 2^x - 1 \quad x \in \mathbb{Z}$.

$$(*) \leq \sum_x p(x) (2^{l(x) - l'(x)} - 1)$$

$$= \sum_x 2^{-l(x)} (2^{l(x) - l'(x)} - 1)$$

$$= \sum_x 2^{-l'(x)} - \sum_x 2^{-l(x)}$$

$\underbrace{\hspace{10em}}_{=1} \quad \underbrace{\hspace{10em}}_{p(x)} \quad \leq \quad \square$

Rmk: " $=$ " $\Leftrightarrow l'(x) = l(x) \quad \forall x \in X$.